

Chapter 10



New AI-Based Trends in Analytics and Data Science

Learning Objectives (1 of 2)

- 10.1** Describe the current and future use of **cloud computing** in business analytics
- 10.2** Describe how **geospatial and location-based analytics** are assisting organizations
- 10.3** Describe **image analytics** and use of images as an alternative source of data
- 10.4** Describe the **IoT**, its characteristics, & the benefits and drivers
- 10.5** Describe typical **IoT applications** in a diversity of fields, smart appliances and homes

Learning Objectives (2 of 2)

- 10.6** Understand the concept of **smart cities**, their content, and their benefits
- 10.7** Understand the potential of using **network properties in analytics**
- 10.8** Describe other **emerging applications in AI**: genomics , health, etc.

How is IoT related to data analytics?

Opening Vignette

Analysis of Sensor Data Helps **Siemens Avoid Train Failures**

Discussion Questions

1. In industrial equipment such as **trains**, what parameters might one measure on a regular basis to estimate **the equipment's current performance** and future repair needs?
2. How would **weather data** be useful in analyzing a train's equipment status?
3. Estimate **how much data** you might collect in one month using, say, 1,000 sensors on a train. Each sensor might yield 1 KB data per second.
4. How would you propose to store such data sets?

Next Case (1 of 2)

“Discover Foods” Employs IoT and Machine Learning to Ensure Food Quality

- Discover Foods is one of market leaders in the food processing industry.
- For Discover Foods, it is very important to **predict the temperature of the food** for maintaining its safety.
- A significant factor in assessing **the food quality based on temperature** is the nature of the food itself.
 - Consider a food that must be maintained at a subzero temperature for optimal quality.
 - The quality of food gets affected when the temperature fluctuates (rises and falls several times) as in a typical temperature time series graph.

Next Case (2 of 2)

- Remote sensing, cloud computing, and machine learning can potentially revolutionize the food industry
- New AI-based technologies can help automate the maintenance of temperature, microorganism growth, humidity and other parameters pertaining to the upkeep of food standards in real time.
- The implementation of AI-based intelligence systems in Food Safety should yield greater food quality, lower operational costs, and minimal losses from the spoilage of edibles.

Cloud Computing

- An emerging and highly promising technology trend in business analytics and data science
- “**Cloud computing**” is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, and services) that can be rapidly provisioned and released with minimal management effort or service-provider interaction.”

Cloud-Based Analytics (1 of 11)

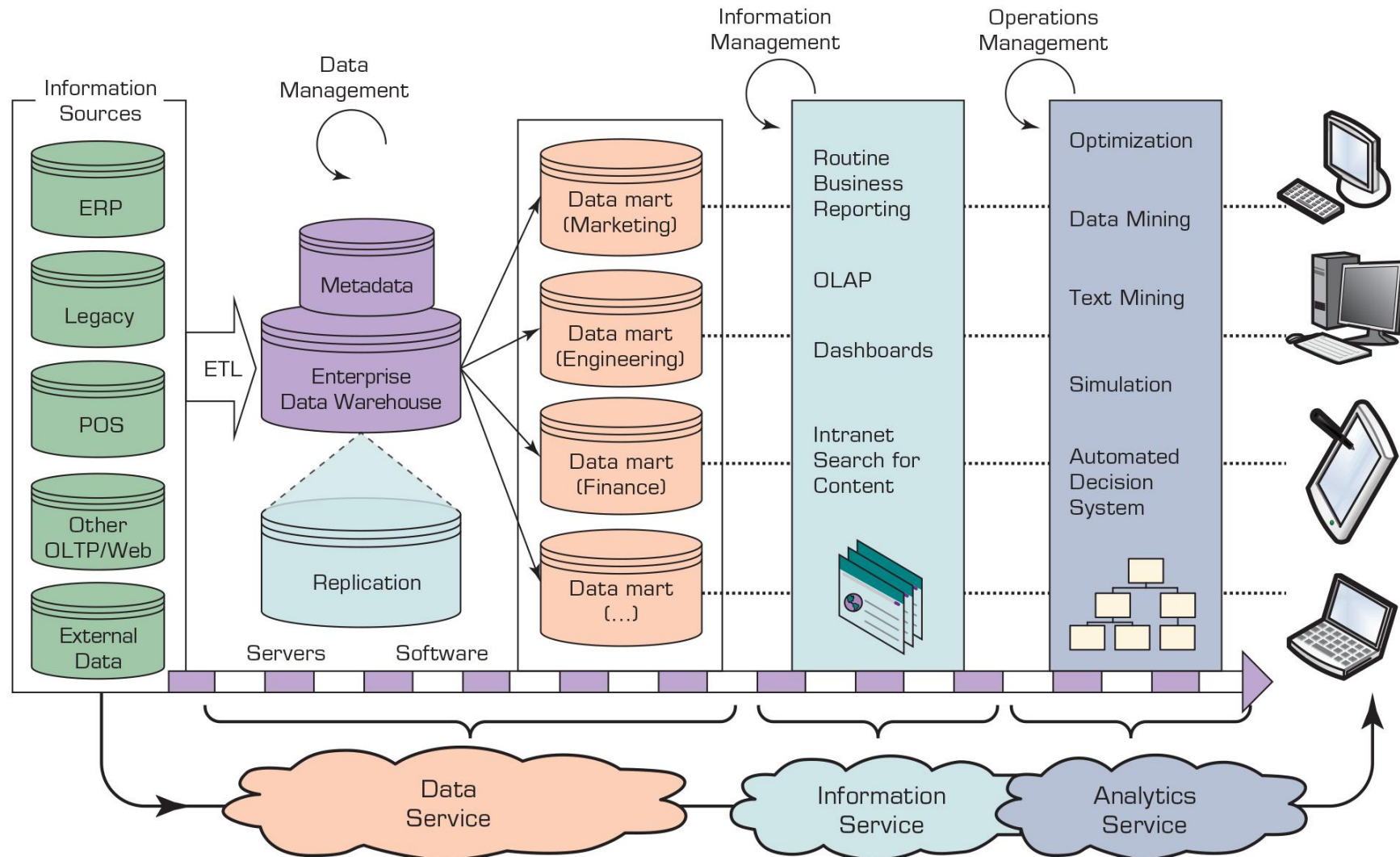
- A new name for the previous related **trends**:
 - Utility computing
 - Application service provider
 - Grid computing
 - On-demand computing
 - Software as a service (SaaS)
 - And even older, centralized computing with dumb terminals
- Cloud computing originates from a reference to the Internet as a “**cloud**”
- It represents an **evolution of all the previously shared/centralized computing trends**

Cloud-Based Analytics (2 of 11)

- Cloud computing examples
 - **Web-based e-mail**. It complies with cloud computing requirements – data storage and processing on the provider supplied software and hardware, centralized control and upgrade, etc.
 - **AWS** is a good general business example of cloud computing is Amazon.com's Web services - an impressive technology infrastructure for e-commerce as well as for BI, customer relationship management, and supply-chain management
 - Others include Salesforce.com, IBM Cloud, Microsoft Azure, Google, Adobe, and many others.
- Cloud computing, like many other IT trends, has resulted in new **offerings in analytics**, e.g., cost effective, scalable data warehousing

Cloud-Based Analytics (3 of 11)

- A conceptual architecture of a cloud-oriented support system



Cloud-Based Analytics (4 of 11)

- A service-oriented decision support environment is essentially a cloud-based analytics system
- In service-oriented decision support solutions, following critical components can be obtained either individually or as a bundled and provided to the users as service:
 1. operational systems
 2. data warehouses
 3. online analytic processing, and
 4. end-user components
- Because the field of cloud computing is fast evolving and growing, there is much confusion about the terminology being used by various vendors and users

Cloud-Based Analytics (5 of 11)

- **Data-as-a-Service (DaaS)**

- Where data lives (the actual platform on which the data resides) does not matter - data can reside in a local computer or in a server
- Any business process can ***access data wherever it resides***
- Data preprocessing and quality assurance happens centrally
- Access to data via open standards such as SQL, X Query, and XML
- Supports agility and flexibility (use of any analytics platforms)

- **Desktop as a Service (DaaS)**

- Providers ***offer virtual desktop*** with all the required applications
- Centralizes the software, hardware, and security of the desktop
- No need to deal with the authorization and complexities of software installation on individual machines (also called P C-as-a-Service)

Cloud-Based Analytics (6 of 11)

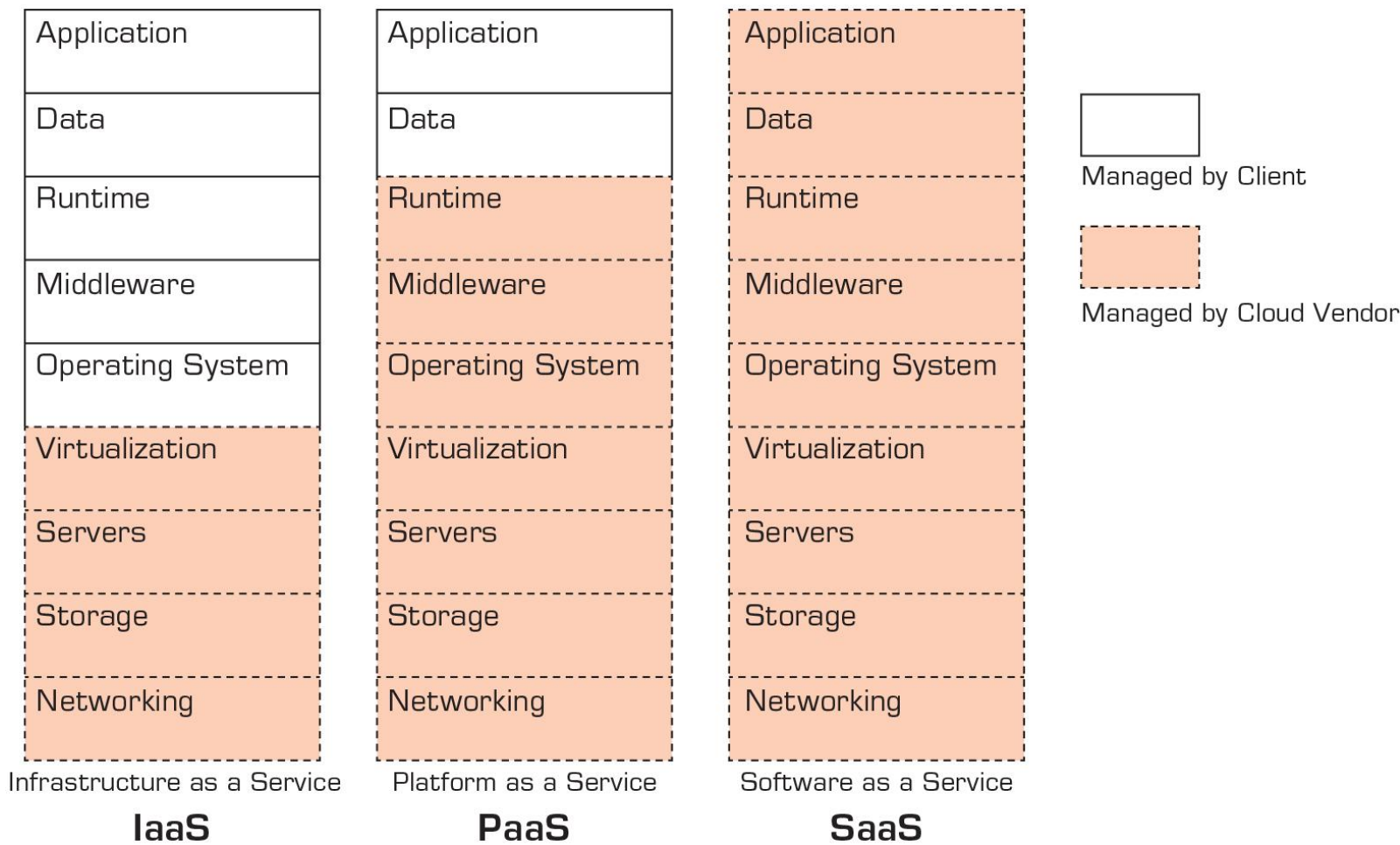
- **Software as a Service (SaaS)**
 - *Use of applications and software* that run on distant computers in some cloud infrastructure
 - Consumers need not worry about managing underlying cloud infrastructure and must pay for the use of software only
 - All we need is a Web browser or an app on a mobile device to connect to the cloud (e.g., Gmail, SAS Viya, etc.)
- **Platform as a Service (PaaS)**
 - Companies can deploy their software and applications in the cloud so that their customers can use them
 - No need to worry about and manage the *underlying resources* (e.g., servers, storage, or operating systems)
 - Allows focusing on their business functions, not on platform

- **Infrastructure as a Service (IaaS)**
 - Infrastructure resources like networks, storage, servers, and other computing resources are provided to client companies
 - A good example of this is Amazon Web Services (AWS)
- Essential Technologies for Cloud Computing - **Virtualization**
 - Creation of a virtual version of operating system or server
 - Virtualization can be in all three areas of computing:
 - **Network virtualization** - splitting of available bandwidth into channels and allocating them to a particular server/device
 - **Storage virtualization** - pooling of physical storage from multiple network storage devices into a single storage device
 - **Server virtualization** - masking of physical servers from server users, thereby centralizing and simplifying management

Cloud-Based Analytics (8 of 11)

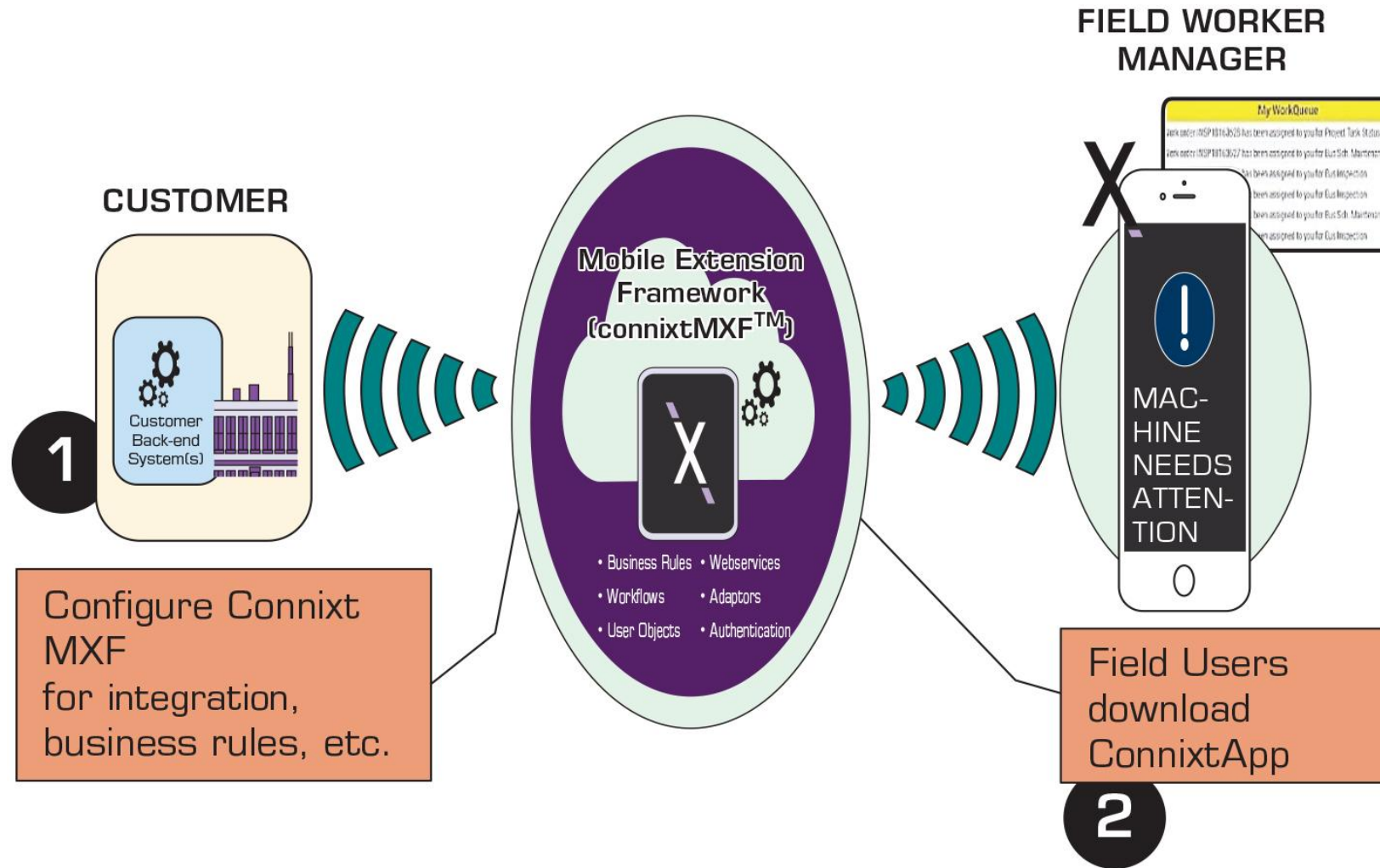
Technology Stack as a Service

Figure 10.2 Technology Stack as a Service for Different Types of Cloud Offerings.



Analytics in Action 10.1 (1 of 2)

Major West Coast Utility Uses Cloud-Mobile Technology to Provide Real-Time Incident Reporting



Analytics in Action 10.1 (2 of 2)

Major West Coast Utility Uses Cloud-Mobile Technology to Provide Real-Time Incident Reporting

Questions for Discussion

1. How does cloud technology impact enterprise software for small and mid-size businesses?
2. What are some of the areas where businesses can use mobile technology?
3. What types of businesses are likely to be the forerunners in adopting cloud-mobile technology?
4. What are the advantages of cloud-based enterprise software instead of the traditional on-premises model?
5. What are the risks of cloud versus traditional on-premises applications?

Cloud-Based Analytics (9 of 11)

- **Cloud Deployment Models**
 - Private cloud – operated solely for and by a single organization
 - Public cloud – shared/offered by service providers over the Internet
 - Hybrid cloud – a mix of private and public clouds
- Major **Cloud Platform Providers** in App Development and Deployment
 - Amazon Elastic Beanstalk
 - IBM Cloud
 - Microsoft Azure
 - Google App Engine
 - OpenShift - Red Hat's cloud application platform based on a PaaS

Cloud-Based Analytics (10 of 11)

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- **Analytics as a Service (AaaS)**
 - It is an extensible analytical platform using a cloud-based delivery model, where various BI and data analytics tools serviced
 - It combines aspects of cloud computing with Big Data analytics and empowers data scientists and analysts
- ***Representative Analytics as a Service Offerings:***
 - IBM CLOUD
 - AMAZON QUICKSIGHT - <https://aws.amazon.com/quicksight/>
 - GOOGLE DATA STUDIO - <https://datastudio.google.com/>
 - MINEMYTEXT.COM - <https://minemytext.com/>
 - SAS VIYA
 - TABLEAU
 - SNOWFLAKE

Illustrative Analytics Applications Employing the Cloud Infrastructure

- Using Azure IoT (Internet of Things), Stream Analytics, and Machine Learning to Improve **Mobile Healthcare Services**
 - Zion China, which is a provider of mobile healthcare services, has produced **an innovative health monitoring tool** about glucose levels, blood pressure, diet, medication, and exercise for their users
 - Uses Stream Analytics, Machine Learning, IOT, and Power BI
- Chime **Enhances Customer Experience** Using Snowflake
 - Chime, a banking option, offers a Visa debit card, FDIC-insured spending and savings account, and a mobile application app that makes banking easier for people
 - Chime uses Snowflake Elastic Data Warehouse solution to pull data from 14 different data sources to enhance member services and provide a more personalized banking experience to customers

Cloud-Based Analytics (11 of 11)

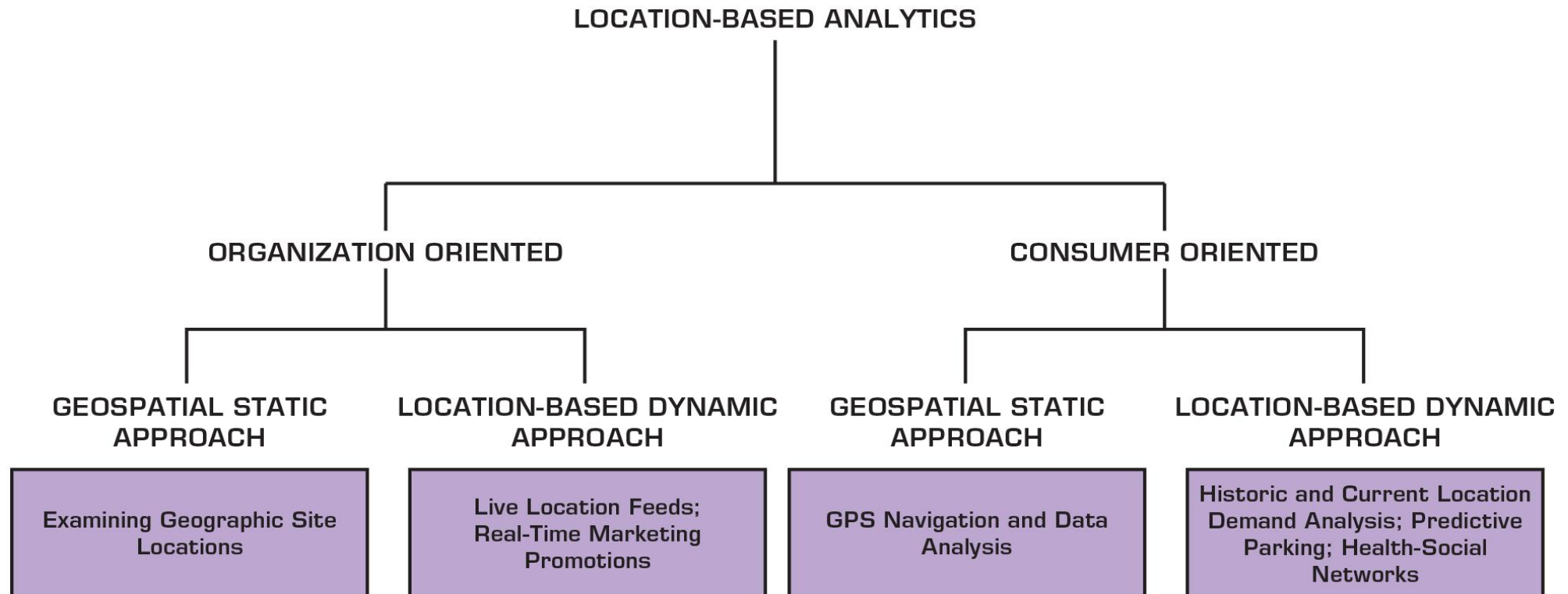
Security Concerns

- As we are entering the “petabyte age,” and traditional data and analytics approaches are beginning to show their limits
 - A major advantage of cloud computing offerings is **the rapid diffusion of advanced analysis tools** among the users, without significant investment in technology acquisition
- Major challenges:
 - Security, service level, and data governance, loss of control and privacy, legal liabilities, cross-border political issues
 - The top three security threats (Cloud Security Alliance):
 - Data loss and leakage
 - Hardware failure of equipment, and
 - Insecure interface

What is location-based
analytics?

Location-Based Analytics (1 of 4)

- Incorporating **location data** in analytics
 - Static (e.g., geospatial data) versus dynamic location data
 - Helping organizations versus for helping individuals/consumers



Location-Based Analytics (2 of 4)

- **Geospatial Analytics**

- Organizations to add a new dimension of “where” to their traditional business analyses (which currently answers the questions of “who,” “what,” “when,” and “how much”)
- Location-based data are now readily available from geographic information systems (GIS)
- A company that is the market leader in providing GIS data is ESRI (esri.com) via their flagship product, Arc GIS
- A few examples:
 - Agricultural applications (proagrica.com and ageagle.com)
 - Crime analysis
 - Disease spread prediction
- Organizations can quickly overlay weather and environmental effects for better and richer forecasting and planning

Analytics in Action 10.3

Improving Home Loan Appraisal Process Using BI and Geographic Information Systems

Questions for Discussion

1. What exactly are the components to think about with respect to **property appraisal**?
2. Put together a **GIS-based business intelligence** design with the house appraisal procedure as well as determine the elements and their related features.
3. Assuming being part of appraisal team for home loan application, what **factors** can be suggested to the house loan committee?

Analytics in Action 10.4

Starbucks Exploits GIS and Analytics to Grow Worldwide

Questions for Discussion

1. What type of demographics and GIS information would be relevant for **deciding on a store location**?
2. It has been mentioned that Starbucks encourages its customers to use its mobile app. What **type of information** might the company gather from the app to help it better plan operations?
3. Will the availability of free Wi-Fi at Starbucks stores provide any information to Starbucks for better analytics?

Location-Based Analytics (3 of 4)

- **Real-time Location Intelligence**

- Many devices (Cars, buses, taxis, mobile phones, cameras, and personal navigation devices, etc.) in use by consumers and professionals are **constantly sending out their location information**
 - Via GPS, Wi-Fi, and cell tower triangulation
- By analyzing the large-scale patterns of movement, it is possible to identify distinct classes of behaviors in specific contexts
 - Allows a businesses to better understand its customer patterns and make more informed decisions about promotions, pricing, etc.
- **Augmented reality** - an extension of location-based analytics
 - Reportedly, a big push at both Microsoft and Apple
 - Technologies depend upon the location information, images through camera, as well as other sensors (when available) to provide the user an immersive experience

Location-Based Analytics (4 of 4)

- **Applications for Consumers**

- Tremendous growth in smartphones/apps creating opportunities where the consumers use analytics without ever realizing it
- These apps are meant for enabling consumers to make better decisions by employing specific location-based analytics
 - **Waze**, a social Web app that assists users in identifying a navigation path and **alerts users about potential issues** such as accidents, police checkpoints, speed traps, and construction, based on other users' inputs
 - **Park PGH**, app deployed in Pittsburgh, PA, developed in collaboration with Carnegie Mellon University, includes predictive capabilities to **estimate parking availability**
- Can create many entrepreneurial opportunities
- **One key concern** in employing these technologies is the loss of privacy

Image Analytics/Alternative Data

- Advances in **image capturing through high-resolution** cameras, storage capabilities, and deep learning algorithms have enabled remarkably interesting analyses
 - Satellite data have often proven their utility in many fields including science, e.g., scientists who need to regularly check global change, land usage, and weather
- This type of Big Data (aka Alternative Data) is used in **many applications**, including:
 - World Bank uses satellite data to propose strategic recommendations for **urban planners** and officials from developing nations
 - EarthCast provides correct **weather updates** for a large commercial U.S. airline based on the data from 60 government-operated satellites
 - Imazon is using satellite data to **detect Amazon deforestation**
 - Global Fishing Watch spots where and when **fishing illegally**

Analytics in Action 10.5

Image Analysis Helps Estimate Plant Cover

Questions for Discussion

1. What is the purpose of knowing how much ground is covered by green foliage on a farm? In a forest?
2. Why would image analysis of foliage through an app be better than a visual check?
3. What other applications of image analysis can you think of?

Essentials of Internet of Things (IoT)

- **IoT** refers to a computerized network that connects many objects (people, animals, devices, sensors, buildings, items) each with embedded microprocessor
- Connections are made wirelessly via Internet
- IoT allows communication and exchange of data among the object and their environment
- Connections are made anytime, anyplace
 - IoT uses ubiquitous computing

Definitions and Characteristics of IoT (1 of 3)

- “**The Internet of Things** means sensors connected to the Internet and behaving in an Internet-like way by making open, ad hoc connections, sharing data freely, and allowing unexpected applications, so computers can understand the world around them and become humanity’s nervous system.”
 - *Kevin Ashton, Creator of the term Internet of Things*
- “**The IoT** is a network of connected computing devices including different types of objects (e.g., digital machines). Each object in the network has a unique identifier (UID), and it is capable of collecting and transferring data automatically across the network.”
 - *Our working definition*

Definitions and Characteristics of IoT (2 of 3)

- IoT allows people and things to interact and communicate at any time, any place, regarding any business topic or service
- **IoT Characteristics** (Miller, 2015):
 - Large numbers of objects (things) can be connected.
 - Each thing has a unique definition/I D (I P address).
 - Each thing can receive, send, and store data automatically.
 - Each thing is delivered mostly over the wireless Internet.
 - Each thing is built upon machine-to-machine communication.
- Internet connects people to each other using computing technology, **while IoT connects “things”** (physical devices and people) to each other and to sensors that collect data

Definitions and Characteristics of IoT (3 of 3)

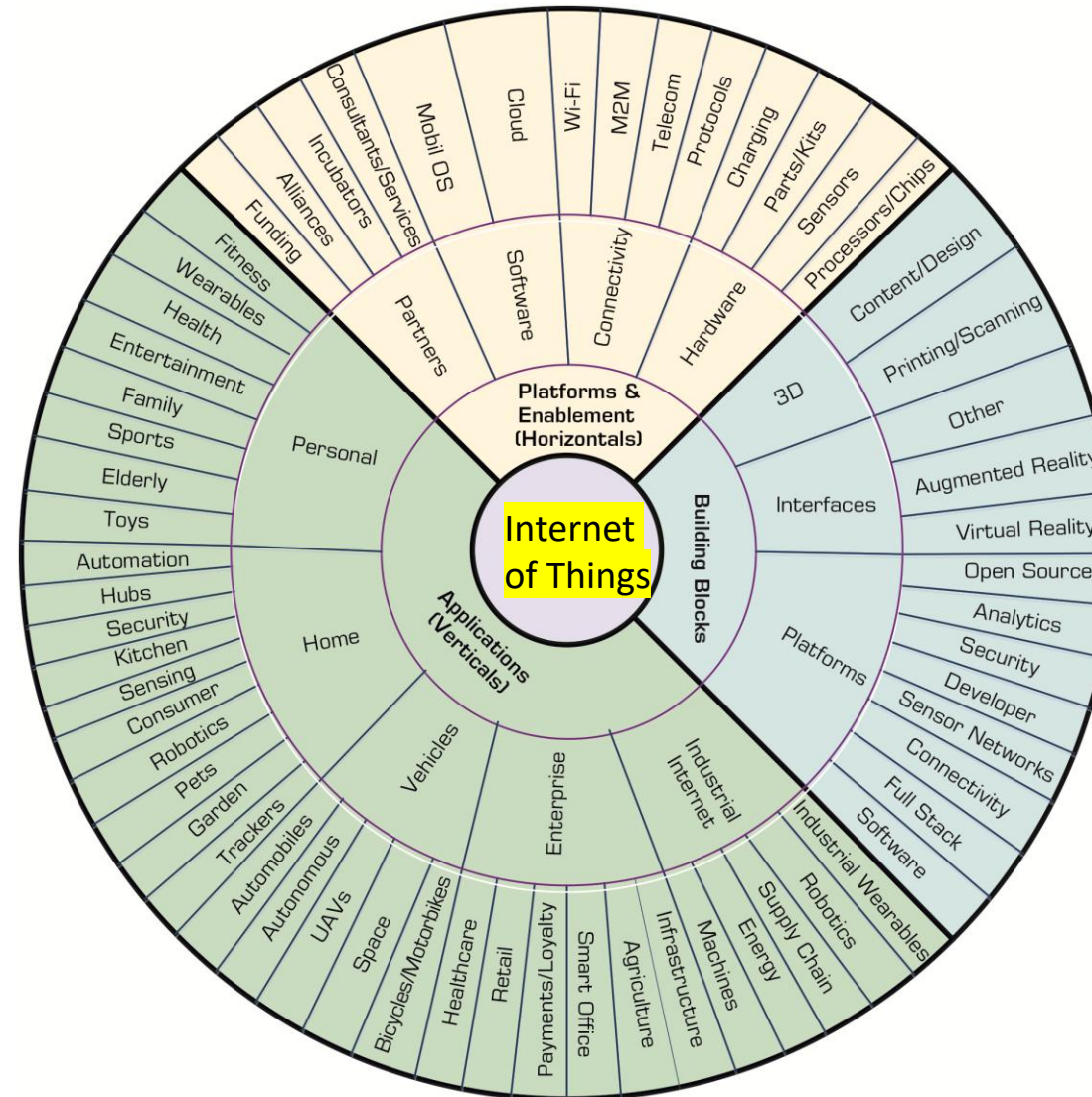
- **Simple Examples**
 - Autonomous vehicles (more on this later)
 - Smartbin (“smart” trash containers)
 - Refrigerator automatically ordering food items
 - Clorox introduced a new Brita filter that automatically orders water filter when needed
- IoT is **Changing Everything** (in production/logistics)
 - Real-time tracking of everything
 - Real-time route tracking, logistics optimization
 - Real-time condition assessment of critical objects
 - Real-time monitoring of the perishable items

IoT Ecosystem (1 of 2)

- The IoT ecosystem refers to all components that enable users to create IoT applications
 - Including gateways, analytics, AI algorithms, servers, data storage, security, and connectivity devices
- Platforms
 - Software, hardware, connectivity, ...
- Building blocks
 - Interfaces, platforms, 3D, ...
- Applications
 - Personal, home, vehicle, industrial, enterprise

IoT Ecosystem (2 of 2)

Figure 10.5 An Illustrative IoT Ecosystem



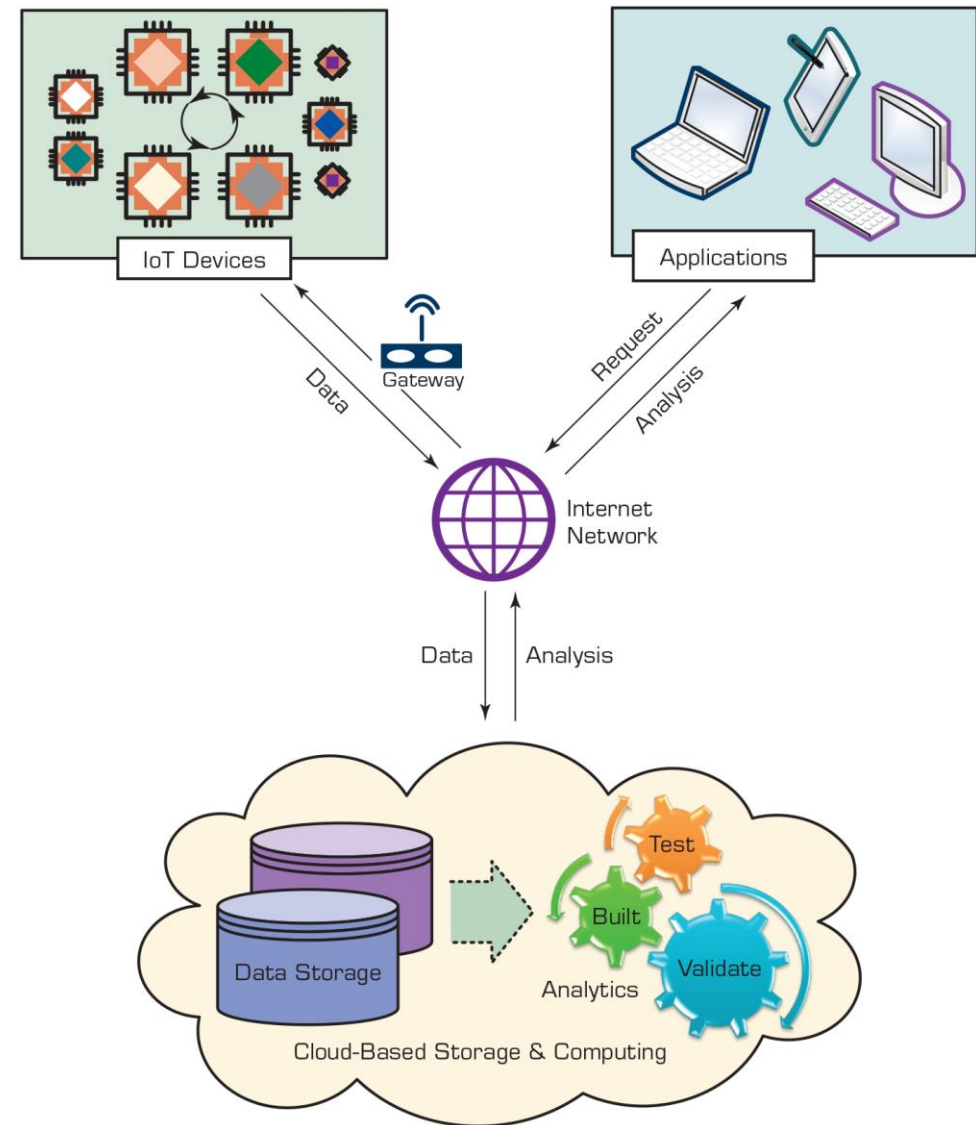
Structure of IoT Systems (1 of 2)

- IoT Technology Infrastructure (four major blocks)
- **Hardware**
 - physical devices, sensors, and actuators where data are produced and recorded
- **Connectivity**
 - Via hubs, gateways and Internet/Cloud)
- **Software backend**
 - The logic/process implementation that manages data, often in the cloud)
- **Applications**
 - The use of the generates data → information for some specific purposes

Structure of IoT Systems (2 of 2)

Figure 10.6 The Building Blocks of IoT.

- Implementations often utilize IoT Platforms
 - Amazon AWS IoT,
 - Microsoft Azure IoT suite,
 - Predix IoT Platform by General Electric (G E),
 - IBM Watson IoT platform
 - Teradata Unified Data Architecture



Major Benefits and Drivers of IoT (1 of 2)

- **Major Benefits of IoT**
 - Reduces cost by automating processes.
 - Improves workers' productivity.
 - Creates new revenue streams.
 - Optimizes asset utilization (e.g., see the opening vignette).
 - Improves sustainability.
 - Changes and improves everything.
 - May anticipate our needs (predictions).
 - Enables insights into broad environments.
 - Enables smarter decisions/purchases.
 - Provides increased accuracy of predictions.
 - Identifies problems quickly (even before they occur).
 - Provides instant information generation and dissemination.
 - Offers quick and inexpensive tracking of activities.

Major Benefits and Drivers of IoT (2 of 2)

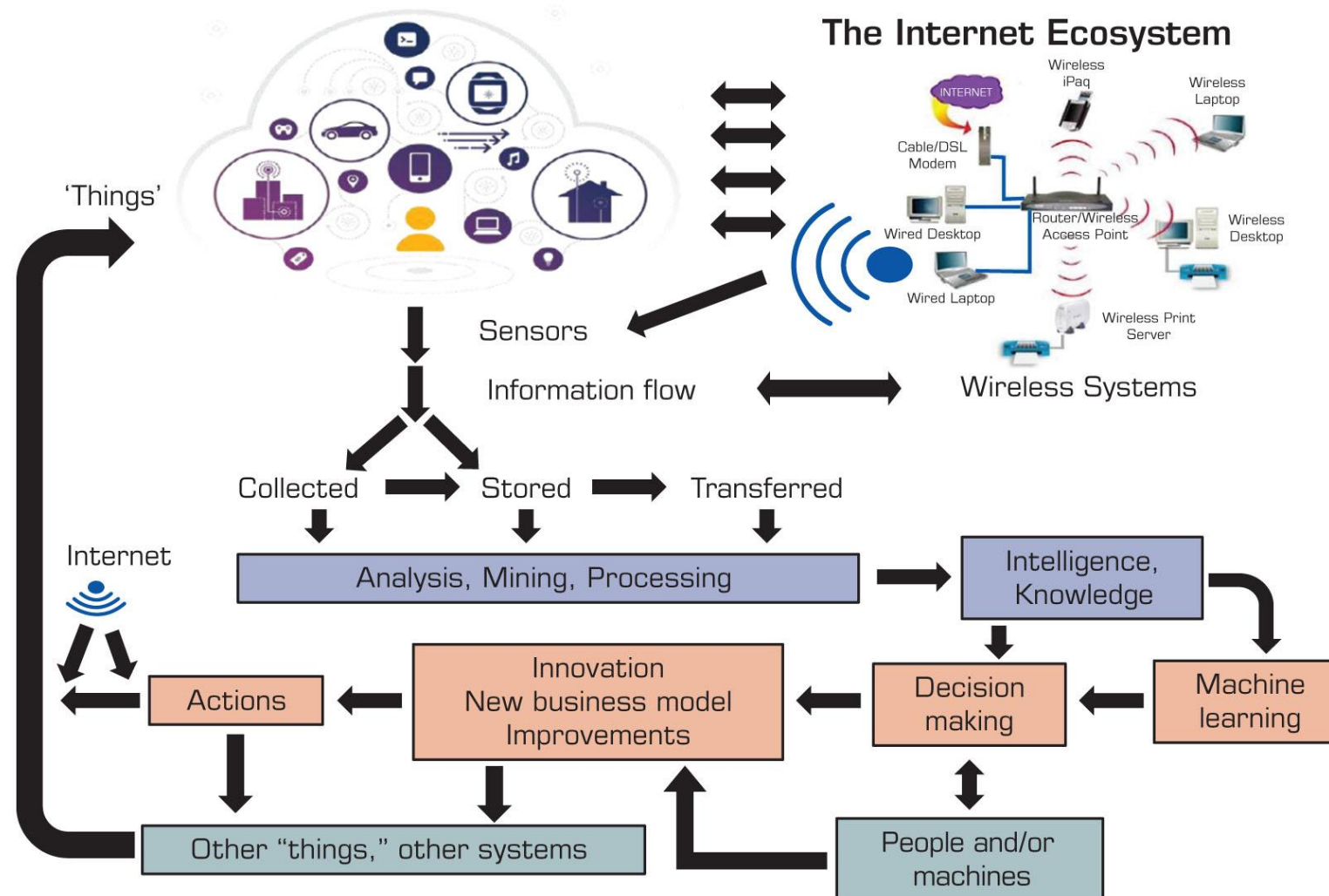
- **Major Drivers of IoT**
 - Very large number of “things” will be connected to the Internet
 - Connected autonomous “things”/systems (e.g., robots, cars) create new IoT applications.
 - Broadband Internet is more widely available, increasing with time.
 - The cost of devices and sensors is continuously declining.
 - The cost of connecting the devices is decreasing.
 - Additional devices are created (via innovations) and are interconnected easily
 - More sensors are built into devices.
 - Smartphones’ penetration is skyrocketing.
 - The availability of wearable devices is increasing.
 - The speed of moving data is increasing to 60H T z.

How IoT Works (1 of 2)

- IoT is not an application.
- It is an infrastructure, platform, or framework that is used to support applications.
- A simple view to how IoT works:
 - The Internet ecosystem includes a large number of things
 - Sensors and other devices collect information from the ecosystem
 - The collected information can be displayed, stored, and processed analytically (e.g., by data mining)
 - This analysis converts the information into knowledge and/or intelligence
 - Expert systems or machine learning may help in turning the knowledge into decision support (made by people and/or machines), which is evidenced by improved actions and results
 - • • leading to new applications and use cases.

How IoT Works (2 of 2)

Figure 10.7 The Process of IoT.



Sensors and Their Role in IoT (1 of 2)

- **A sensor** is an electronic device that automatically collects data about events in its environment
- Sensors are a critical part of IoT systems
- There are several types of sensors and several methods for collecting data from sensors
- Sensors are also essential components in robotics and autonomous vehicles
- Sensors have distance and precision limits
 - Sensors of a very short range are known as *proximity sensors*, which are more reliable than longer ranges

Sensors and Their Role in IoT (2 of 2)

- **How Sensors Work with IoT**
 - In large-scale applications, sensors collect data that are transferred to processing in the “cloud”
- Sensor Applications and Radio-Frequency Identification (**RFID**) Sensors
 - Sensors can measure many things: humidity, temperature, etc.
 - A well-known type of sensor that plays an important role in IoT is radio-frequency identification
- RFID in conjunction with other sensors play a major role in IoT applications

Analytics in Action 10.7

Rockwell Automation Monitors Expensive Oil and Gas Exploration Assets to Predict Failures

Questions for Discussion

1. What type of information would likely be collected by an oil and gas drilling platform?
2. Does this application fit the three Versus (volume, variety, velocity) of Big Data? Why or why not?

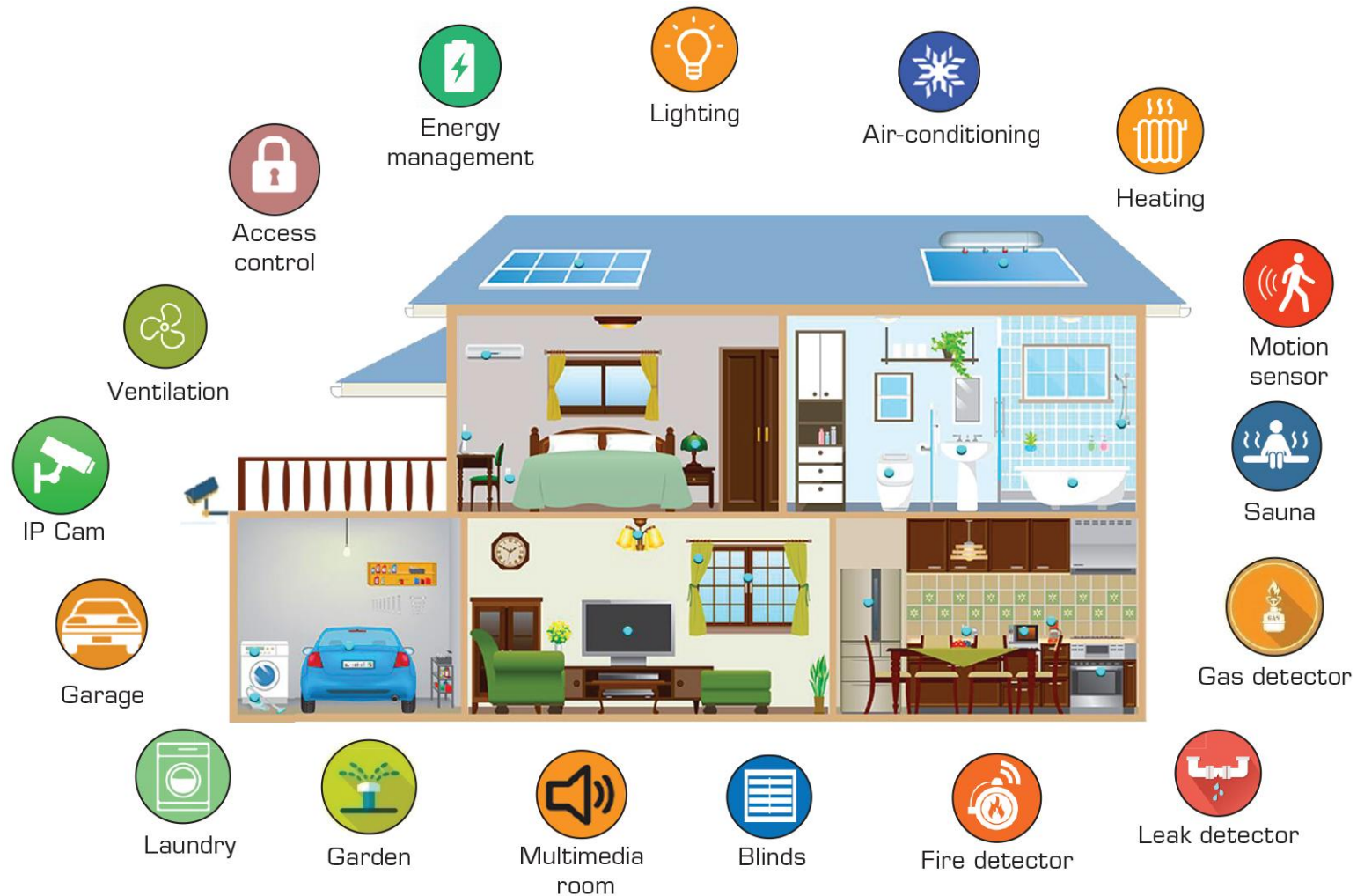
Technology Insight 10.1

RFID Sensors

- RFID - a generic technology that uses of **radio-frequency waves** to identify objects
- Part of a family of automatic identification technologies that also includes ubiquitous barcodes and magnetic strips
 - RFID stores richer identification data
- Use of RFID spread by retailers' supply-chains
- RFID works with tags and readers
 - Active vs passive tags (long/short range)

IoT Applications (1 of 5)

Figure 10. 8 The Components of a Smart Home.



IoT Applications (2 of 5)

- French National **Railway System's** Use of IoT
 - Manage 15,000 trains and 30,000 kilometers
 - Provide quality, availability, and safety for its nearly 14 million passengers
 - Uses IBM Watson in the background
- **Smart Homes and Appliances**
 - A home with automated components that are interconnected such as lights, appliances, security, and entertainment
 - Designed to provide their dwellers with comfort, security, low energy cost, and convenience
 - Most existing home are not smart, but they can inexpensively be equipped with partial smartness
 - See https://techterms.com/definition/smart_home

IoT Applications (3 of 5)

- **Smart Appliances** (for Smart Homes)
 - A smart appliance includes features that can remotely control the appliance operations, based on the user preferences.
 - Some common products that help make a home a smart home include:
 - Learning thermostat
 - Smoke detector and alarm
 - Video surveillance systems
 - Samsung Smart fridge
 - 10 in 1 device for the kitchen
 - Instaview Refrigerator from LG
 - Whirlpool's smart top load washer

IoT Applications (4 of 5)

- **Smart Homes**

- A Smart Home Is Where the Bot Is
 - AI based bots in smart homes with predictive abilities
- Barriers to Smart Home Adoption
 - **Compatibility.** Too many vendors and products; connecting them all is a challenge
 - **Communication.** Different consumers have different ideas on what the smart home should be
 - **Concentration.** Brands need to concentrate on population segments that are most interested in smart homes (e.g., Gen Y)
 - Also - cost justification, invasion of privacy, security, and ease of use

IoT Applications (5 of 5)

- **Smart Components** in Smart Cities and Smart Factories
 - **Smart Digital Factories**
 - A flexible system that can self-optimize performance across a broader network, self-adapt to and learn from new conditions in real or near real time, and autonomously run entire production processes.
 - **Improving Transportation in a Smart City**
 - A major area of improvement in a smart city is transportation (i.e., increased number of vehicles and the inability to accommodate all of them effectively). Examples include
 - Valerann, an Israeli start-up, developed smart road studs
 - Smart Mobility Consortium (Hong Kong) works on mobility in the smart city of Hong Kong

Autonomous (Self-Driving) Vehicles

- **Autonomous vehicles** = driverless cars, robot-driven cars, self-driving cars, etc.
 - Google started it, Tesla made it famous
 - Most car manufacturers are investing in this technology
 - France, Singapore, China, and several other countries already have these cars
- Implementation Issues in Autonomous Vehicles
 - The cost of real-time 3D map technologies
 - AI software must be nimble, and its capabilities increased
 - Readyng the automakers and insurers
 - Sensor technology needs improvement and cost efficiency
 - IoT systems and connectivity need to be improved

5G Technologies and Impact on AI

- Fifth-generation (5G) mobile technology is being touted as the catalyst for the Fourth Industrial Revolution
 - 5G's superior speeds and ultra-low latencies should enable diverse set of breakthrough use cases for consumers and businesses
- Key advantages and benefits of 5G technology
 1. Provides very high-speed access, even in densely populated areas
 2. Connects everything, thereby supporting massive IoT and machine-to-machine (M2M) communications
 3. Provides real-time latency, thereby minimizing delays in network response time and enabling entirely new services and applications for vertical industries
 4. Delivers significant improvements to security and privacy
 5. Provides a service-based architecture (SBA) that delivers increased flexibility and service diversity

Other Emerging AI Topics:

Robotic Process Automation (RPA)

- **RPA**: a technology which reduces the workloads of the employees by automating the repetitive, high-volume steps in processes
 - **Software robots or “bots”** replicate the actions of human workers
- By handling the manual work of humans, RPA gives more time for humans to spend on other important activities
 - RPA can help achieve an increase in productivity, elimination of human errors, lower operational cost, automation of labor-intensive tasks, and faster completion of work, which saves time and money

Bioinformatics and Network Science

- **Bioinformatics** in general and health analytics in particular have emerged as major areas of growth in AI
- **Network Analysis** (NA) is a popular method for analyzing complex problems involving interactions among the features or the observations
 - Network analytics techniques used in conjunction with other approaches such as deep learning or natural language processing
 - Example: comorbidity networks for health disparity analysis
- Network analysis is a popular research area for all three analytics echelons: descriptive, predictive and prescriptive
- As everything is becoming connected, networks science is gaining popularity as an analytics method

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Analytics in Action 10.11

Network analytics for predictive modeling: Predicting hospital length of stay using comorbidity networks

Questions for Discussion

1. What was the challenge?
2. What was the proposed solution?
3. What were the obtained results?

Other Recent Developments

- Web 3.0 – decentralized structure
 - Blockchain, cryptocurrency, non-fungible tokens (NFT)
- Metaverse and Digital Twins
 - **Metaverse** - a virtual replica of a physical object or system objects interact with each other
 - **Digital twin** - a digital representation of an intended or actual real-world physical product, system, or process that serves as the effectively indistinguishable digital counterpart
- GPT-3/Chat GPT
- LaMDA – (Language Model for Dialogue Applications) is Google's adaption of technologies similar to GPT-3
- Blenderbot – Meta/Facebook's AI-based chatbot